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Assignment 201

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IARCO RESEARCH PROPOSAL

- Full Name: Khondaker Md Ziaul Hassan
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- Class/Grade/Year: Year 1
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- Submission Date: September 20, 2026

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- **Research Topic: The Impacts of Generative AI Dependency vs Cognitive Retention among Private STEM Engineering Undergraduates.**

Research Title:

Generative AI Dependency vs. Cognitive Retention: An Empirical Study of Private STEM Engineering Students in Bangladesh

Generative AI Dependency vs. Cognitive Retention: An Empirical Study of Private STEM Engineering Students in Bangladesh

I. INTRODUCTION

A. Background of the Study

12 Generative AI (GenAI) tools such as Claude, Gemini & ChatGPT have increasingly become the norm in daily academic life. While these tools undoubtedly have benefits, the long-term concerns outweigh the short-term gains due to overreliance on GenAI, particularly for STEM students, where problem-solving ability is one of the core skills trained in the general undergraduate STEM engineering curricula.

B. Literature Review

Observational metrics showed only moderate student familiarity with GenAI writing tools, alongside manageable awareness of misinformation and data security [1]. Simultaneously, when generative AI (GenAI) produces academic writing, discord has materialised over the authorship and ownership of that writing [2]. A large-scale field study confirmed that GenAI remains dominated by perception and attitude studies rather than empirical outcomes [3]. Policy guidance calls for a human-centred, ethically grounded approach to GenAI in the educational sector overall [4]. Quantitative descriptive research has generally shown a positive trend in student attitudes toward the use of generative AI [5]. Additional research emphasises the ongoing challenge of balancing educational innovation with academic integrity [6]. Ultimately, systematic reviews highlight the pressing need for strategic research frameworks to address Generative AI governance in the higher education sector [7].

C. Research Gap

Although global literature investigates general AI adoption and integration, none of the cited articles measures the actual impact specifically on STEM undergraduates. Fundamental concerns and ethical debates are well documented [1], [2], [4], [5], [6], [7], and field studies have shown to favour cognition over assigned parameters [3], but whether daily GenAI dependency diminishes long-term cognitive retention and problem-solving skills in STEM students remains unexplored. This gap is most pronounced in STEM engineering undergraduate education, where problem-solving is the core primary skill taught, and in the emerging context of Bangladesh, it is missing from the cited articles, validating this study.

D. Research Topic

5 This research proposal aims to investigate the correlation between daily usage of Generative AI (GenAI) tools (e.g., ChatGPT, Gemini, Copilot) and long-term cognitive retention of fundamental concepts among private university STEM engineering undergraduates in Bangladesh. Specifically, it evaluates whether frequent AI-generated outputs, such as program code, analytical explanations, practical field deployments, or mathematical solutions, degrade independent problem-solving skills, which are essentially the core learned

skills for STEM engineering students, during closed-book evaluations. Furthermore, the research proposal presents frameworks to reduce cognitive offloading and preserve academic integrity.

E. Research Question

Does frequent reliance on generative AI tools (e.g., ChatGPT, Claude, Gemini) for coursework significantly reduce independent problem-solving ability and cognitive retention among Bangladeshi private STEM engineering undergraduates?

II. METHODOLOGY

This empirical study will undertake a cross-sectional, quantitative survey design to investigate the relationship between Gen AI tool reliance and cognitive retention among private STEM engineering undergraduates in Bangladesh.

A. Sampling & Participants:

Data will be collected from $N = 1100$ undergraduate engineering students (100 students from each B.Sc. engineering discipline) selected via proportional random sampling across 10 top-ranked STEM private engineering universities currently in Bangladesh. The target population spans key undergraduate (B.Sc.) disciplines like: Computer Science & Engineering (CSE), Electrical & Electronic Engineering (EEE), Electronic & Communication Engineering (ECE), Electronics & Telecom Engineering (ETE), Industrial & Production Engineering (IPE), Civil & Environmental Engineering (CEE), Civil Engineering (CE), Software Engineering (SWE), Mechanical Engineering (ME), Biomedical Engineering (BME), Textile Engineering (TE), and Other STEM Engineering majors.

B. Data Collection Instrument:

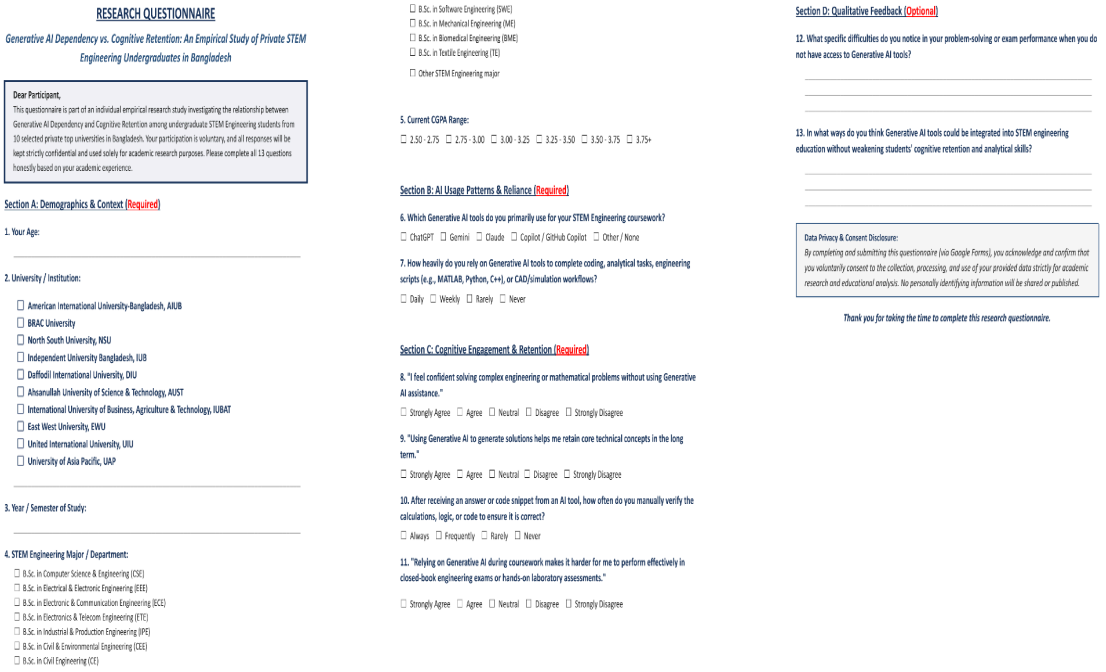
A structured 13-question survey questionnaire, distributed via Google Forms, organised and divided into four primary domains, will be administered:

- Demographics & Academic Context (**required section**): Measures age, institution (from given list), department (from given list), current semester, and CGPA range (from given range).
- AI Usage Patterns & Reliance (**required section**): Evaluates primary GenAI tools used (e.g., ChatGPT, Gemini, Claude, Copilot) and frequency of use across coding, mathematical, analytical, and simulation tasks.
- Cognitive Retention & Confidence (**required section**): Utilises 5-point Likert scales (Strongly Agree to = Strongly Disagree) to measure independent problem-solving confidence, self-perceived long-term concept retention, frequency of manual verification, and exam performance degradation.
- Qualitative Feedback (**optional section**): Captures open-ended insights on specific problem-solving barriers experienced without AI tools and strategies for balanced pedagogical integration.

C. Data Analysis:

16 Quantitative survey data will be processed using IBM SPSS Statistics software. Descriptive statistics (frequencies, percentages, means) will summarise participant demographics, Gen AI tool usage, and reliance patterns. Chi-Square ($\mathbf{X^2}$) tests will quantify variance in AI reliance across different B.Sc. disciplines and CGPA ranges (Section A of the questionnaire). Statistical Association and Ordinal Modeling will be performed in SPSS to measure the direct relationship between Gen AI dependency (Section B of the questionnaire) and student problem-solving confidence, verification habits, and closed-book exam performance (Section C of the questionnaire). Qualitative responses from Section D, the optional section of the questionnaire, will be evaluated using thematic analysis.

13-question survey questionnaire to be distributed via Google Forms, horizontal, sideways-stacked PNG file.



RESEARCH QUESTIONNAIRE

Generative AI Dependency vs. Cognitive Retention: An Empirical Study of Private STEM Engineering Undergraduates in Bangladesh

Dear Participant,
This questionnaire is part of an individual empirical research study investigating the relationship between Generative AI Dependency and Cognitive Retention among undergraduate STEM Engineering students from 10 selected private top universities in Bangladesh. Your participation is voluntary, and all responses will be kept strictly confidential and used solely for academic research purposes. Please complete all 13 questions honestly based on your academic experience.

Section A: Demographics & Context (Required)

1. Your Age:

2. University / Institution:

- ☐ American International University-Bangladesh, AIUB
- ☐ BRAC University
- ☐ North South University, NSU
- ☐ Independent University Bangladesh, IUB
- ☐ Daffodil International University, DIU
- ☐ Ahsanullah University of Science & Technology, AUST
- ☐ International University of Business, Agriculture & Technology, IUBAT
- ☐ East West University, EWU
- ☐ United International University, UIU
- ☐ University of Asia Pacific, UAP

3. Year / Semester of Study:

4. STEM Engineering Major / Department:

- ☐ B.Sc. in Computer Science & Engineering (CSE)
- ☐ B.Sc. in Electrical & Electronic Engineering (EEE)
- ☐ B.Sc. in Electronic & Communication Engineering (ECE)
- ☐ B.Sc. in Electronics & Telecom Engineering (ETE)
- ☐ B.Sc. in Industrial & Production Engineering (IPE)
- ☐ B.Sc. in Civil & Environmental Engineering (CEE)
- ☐ B.Sc. in Civil Engineering (CE)
- ☐ B.Sc. in Software Engineering (SWE)
- ☐ B.Sc. in Mechanical Engineering (ME)
- ☐ B.Sc. in Biomedical Engineering (BME)
- ☐ B.Sc. in Textile Engineering (TE)
- ☐ Other STEM Engineering major

5. Current CGPA Range:

☐ 2.50 - 2.75 ☐ 2.75 - 3.00 ☐ 3.00 - 3.25 ☐ 3.25 - 3.50 ☐ 3.50 - 3.75 ☐ 3.75+

Section B: AI Usage Patterns & Reliance (Required)

6. Which Generative AI tools do you primarily use for your STEM Engineering coursework?

☐ ChatGPT ☐ Gemini ☐ Claude ☐ Copilot / GitHub Copilot ☐ Other / None

7. How heavily do you rely on Generative AI tools to complete coding, analytical tasks, engineering scripts (e.g., MATLAB, Python, C++), or CAD/simulation workflows?

☐ Daily ☐ Weekly ☐ Rarely ☐ Never

Section C: Cognitive Engagement & Retention (Required)

8. "I feel confident solving complex engineering or mathematical problems without using Generative AI assistance."

☐ Strongly Agree ☐ Agree ☐ Neutral ☐ Disagree ☐ Strongly Disagree

9. "Using Generative AI to generate solutions helps me retain core technical concepts in the long term."

☐ Strongly Agree ☐ Agree ☐ Neutral ☐ Disagree ☐ Strongly Disagree

10. After receiving an answer or code snippet from an AI tool, how often do you manually verify the calculations, logic, or code to ensure it is correct?

☐ Always ☐ Frequently ☐ Rarely ☐ Never

11. "Relying on Generative AI during coursework makes it harder for me to perform effectively in closed-book engineering exams or hands-on laboratory assessments."

☐ Strongly Agree ☐ Agree ☐ Neutral ☐ Disagree ☐ Strongly Disagree

Section D: Qualitative Feedback (Optional)

12. What specific difficulties do you notice in your problem-solving or exam performance when you do not have access to Generative AI tools?

13. In what ways do you think Generative AI tools could be integrated into STEM engineering education without weakening students' cognitive retention and analytical skills?

Data Privacy & Consent Disclosure:
By completing and submitting this questionnaire (via Google Forms), you acknowledge and confirm that you voluntarily consent to the collection, processing, and use of your provided data strictly for academic research and educational analysis. No personally identifying information will be shared or published.

Thank you for taking the time to complete this research questionnaire.

D. Data Collection:

Data will be collected over a two-month period to obtain responses from at least $N = 1100$ (100 students from each B.Sc. engineering discipline mentioned in the questionnaire) private STEM engineering undergraduates across the top 10 ranked selected universities in Bangladesh. Direct outreach through digital survey channels, campus networks, and student groups on social media sites will ensure target reach. Strict ethical standards will be enforced; informed consent will be secured from all participants prior to the Google Form submission, ensuring participant anonymity and secure storage strictly for academic evaluation.

E. Potential Limitations

Potential challenges might include skewed sampling across different B.Sc. disciplines, potentially constraining the generalizability of the findings. Additionally, the self-reporting bias could lead participants to downplay AI reliance or overemphasise independent problem-solving confidence due to social desirability. Furthermore, questionnaire fatigue might result in non-response or low completion rates. Ultimately, the cross-sectional design measures self-perceived retention at a single time point, as the survey data is only recorded once, thus omitting long-term cognitive trends.

III. CONCLUSION

This research proposal establishes an empirical framework to evaluate the impact of Generative AI dependency on cognitive retention and independent problem-solving among private STEM engineering undergraduates in Bangladesh. By gathering quantitative questionnaire data from $N = 1100$ students across 10 top private STEM engineering universities in Bangladesh, the study addresses a critical gap in regional educational literature. This study provides empirical insights to guide balanced GenAI integration in STEM engineering curricula while preserving core engineering and analytical capabilities.

IV. FUTURE RESEARCH SCOPE

Future studies can extend this empirical framework by undertaking longitudinal cohort tracking across four-year B.Sc. engineering programs to evaluate actual cognitive retention instead of a single cross-sectional survey completion. Subsequent studies can expand sampling to public STEM institutions in Bangladesh and comparative South Asian universities to assess institutional differences. To mitigate response bias, future frameworks should integrate objective performance metrics, such as non-Gen AI-assisted settings for laboratory examinations and code execution tasks, alongside self-reported data.

Referenced Articles

[1] A.-M. M. Gasaymeh, M. A. Beirat, and A. A. A. Abu Qbeita, "University students' insights of generative artificial intelligence (AI) writing tools," Educ. Sci., vol. 14, no. 10, Art. no. 1062, Sep. 2024, doi: 10.3390/educsci14101062.

[2] A. Bozkurt, "GenAI et al.: Co-creation, authorship, ownership, academic ethics and integrity in a time of generative AI," Open Praxis, vol. 16, no. 1, pp. 5–16, 2024, doi: 10.5598/openpraxis.16.1.628.

[3] L. Law, "Generative Artificial Intelligence (GenAI) in language teaching and learning: A scoping literature review," Comput. Educ.: Artif. Intell., vol. 6, Art. no. 100212, 2024.

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[7] K. Bittle and O. El-Gayar, "Generative AI and academic integrity in higher education: A systematic review and research agenda," in Proc. 57th Hawaii Int. Conf. Syst. Sci. (HICSS), Honolulu, HI, USA, 2024, pp. 1441–1450.